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The impact of fuel composition on the power output and emissions of a gas turbine for gas-to-wire applications

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HIGHLIGHTS

- Use of contaminated associated gas as fuel in power plants.
- Thermodynamic, exergy, and off-design modeling of gas turbines in gas-to-wire systems.
- Parametric analysis of the influence of the CO_2 content mixed with natural gas.
- Relationship between fuel composition and gas turbine performance.

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ABSTRACT

A model of a gas cycle turbine operating under design and off-design conditions with variable fuel compositions for Gas-to-wire (GtW) applications is presented. The model considers the presence of different fractions of combustible (i.e., CH_4, C_2H_6, and C_3H_8) and incombustible (i.e., CO_2) components, while accounting for the system's power and emissions outputs. The content of CO_2 varied from 2 % to 42 %, while the contents of C_2H_6 and C_3H_8 were kept constant at 4 %, all on a molar basis, for all compositions evaluated. To evaluate the behavior of the turbine, its load was varied from 40 % to 100 % of the nominal power. The results obtained indicate that the mass flow rate of both the fuel mixture and the exhaust gases increases with the CO_2 fraction and with the turbine load. The destruction of exergy in the components was presented as a function of the Wobbe index. As the index increases, which is associated with smaller CO_2 fractions, the turbine and compressor exergy show a small reduction, while the destruction in the combustor increases drastically. The addition of CO_2 also significantly influences NO_X and CO emissions. NO_X emissions decrease and CO emissions increase with the reduction of CO_2 present in the mixture. Overall, the results suggest that while more fuel is used to produce the same power with mixtures rich in CO_2, more energy is required to separate CO_2 from combustible fuel, making GtW an alternative to oil wells that produce large amounts of CO_2.

1. Introduction

The development of non-polluting energy sources has become a global driving force for research. In this context, natural gas has gained focus in recent years because renewable sources still face problems with respect to intermittency, requiring the use of fossil fuels as base energy vectors. Thus, natural gas is considered a strategic source, as it is a transition fuel between fossil and renewable energy sources, belonging to the class of *cleaner fuels*, because its combustion emits about 50 % less carbon dioxide (CO_2) compared to other fuels [1,2]. When extracted on offshore oil platforms, natural gas is often associated with oil and water or other impurities, such as H_2S and CO_2, making its use unviable due to the high cost of purity required to meet the requirements for transport through pipelines. For instance, according to [3], the excess gas often present in oil wells is locally burned through a process called *flaring*, which is common during oil and gas processing despite its undesirable emissions.

Studies show that there are approximately 74 trillion cubic meters of associated gas – natural gas produced through oil wells – around the world [4] (apud [5]), although some gas reserves have not been explored yet because they do not provide an adequate financial return. In addition, according to [6], around 2.9 Gm^3 of natural gas are burned

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Nomenclature

a, b, c	chemical reaction coefficients [-]
c_i	constant [-]
CO	CO emission [g/kg$_{fuel}$]
c_p	specific heat capacity at constant pressure [kJ/kg · K]
ex_{air}	air excess [-]
f_m	mass fraction [-]
G	Gibbs free energy [kJ/kmol]
$\dot{V}$	ratio of reduced mass flow rate to its design value [-]
h	enthalpy [kJ/kg]
h_c	enthalpy of combustion [kJ/kmol]
HHV	high heat value [kJ/kg]
HT	heat rate [kJ/kWh]
LHV	low heat value [kJ/kg]
$\dot{m}$	mass flow rate [kg/s]
M	molar mass [kg/kmol]
n	rotating speed [rpm]
n_m	number of moles [-]
NO_X	NO_X emission [g/kg$_{fuel}$]
P	pressure [Pa]
R	universal gas constant [kJ/kmol·K]
T	temperature [K]
$\dot{W}$	power output [kW]
Wo	Wobbe index [kJ/kg]
y	molar fraction [-]
$\dot{X}$	exergy destruction rate [kW]
x	specific exergy [kJ/kg]

Greek symbols

η	efficiency [-]
$\dot{\mu}$	mass flow rate ratio [-]
$\dot{\phi}$	pressure recovery coefficient [-]
π	pressure ratio [K]
ρ	density [kg/m^3]
τ	residence time [s]
θ	fugacity coefficient [-]

Superscripts

$\bar{}$	normalized by design value
$\dot{}$	per unit of time
$\ddot{}$	fraction of the design value
f	physical
q	chemical

Subscripts

1,2,…,k	streams
af	adiabatic flame
c	compressor
comb	combustor
des	design value
f	fuel
gen	electric generator
gtc	gas turbine cycle
pz	primary zone
t	turbine

annually, suggesting that this less polluting fossil-based energy source might be relied on for several years to come. In order to make use of this energy source, Gas-to-wire (GtW) platforms are an alternative. This type of system uses gas to produce electricity at the extraction site, and excess electricity is then transmitted through submarine cables in offshore plants [7]. Gas-to-wire systems are generally more efficient and less costly than standard systems since there is no need to treat and separate the contaminants present in the gas to the levels required for commercialization. Obviously, the turbine specifications must be in accordance with the fuel mixture used.

Not surprisingly, the use of GtW systems has been considered in the specialized literature, including the pre-salt fields present in Brazil. For instance, Relva et al. [5] presented a case study focused on the pre-salt layer, suggesting that the installation of GtW thermoelectric plants is feasible, but some changes must be made to make this possible (e.g., prioritized dispatch contracts). Ref. [8] reviewed eight GtW projects in Asia, Africa, and South America. According to the study, natural gas is attractive as it reduces dependence on oil and pollutes less. Thus, GtW systems might be more interesting in remote regions, as they are an efficient way to generate energy on-site.

Despite the fact that flaring is avoided, controlling CO_2 emissions in GtW systems is still necessary. For example, exhaust gases rich in CO_2 might be captured by a CO_2 capture system and later reinjected into the well, executing the so-called Enhanced Oil Recovery (EOR) technique, which significantly increases well production [5]. However, it should be noted that even with recent research, many points need to be addressed in GtW systems. This is because most available studies exclusively analyze the technical and economic implications of installing GtW systems for power generation, e.g., [6,9,10], while disregarding the presence of contaminants and their effects. Nevertheless, for contextualization purposes, some of the key studies considering associated gas and/or contaminated fuel are reviewed next.

Ref. [11] focused on the use of GtW systems to take advantage of the associated gas, rather than the natural gas, burned in the process. They found that neither the thermal efficiency nor the output power is significantly affected by the use of associated gas. However, more associated gas is consumed to produce the same amount of power. Following this line of research, Somehsaraei et al. [12] analyzed the flexibilization of the fuel used in microturbines (MGTs). The results indicated that the use of biogas instead of natural gas causes an increase in fuel mass flow. At the same time, electrical efficiency decreases with increasing methane content. It should be mentioned that pollutant emissions are also influenced by the fuel composition.

Ref. [13] performed a parametric analysis evaluating the impact of constituents such as ethane (C_2H_6), propane (C_3H_8), isobutane (C_4H_{10}), CO_2, and nitrogen (N_2) on the combustion of methane (CH_4). The results showed that by individually varying the content of constituents such as ethane and propane from 0 to 30 %, the adiabatic flame temperature increases, leading to an increase in the emission of NO_X and CO. At the same time, a decrease in the contents of CO_2 and N_2, decreases the adiabatic flame temperature and, consequently, the emissions of NO_X and CO. Ref. [14] performed a partial load analysis, in addition to verifying the influence of the presence of N_2 in natural gas on the main parameters of the cycle. The study concluded that by increasing the fraction of N_2, the fuel flow rate drops significantly, causing a minimal increase in the first and second law efficiencies. Ref. [15] studied the behavior of a gas turbine powered by a mixture of CH_4 and H_2. It was found that for the same turbine inlet temperature, the efficiencies of the combined cycles considered increased slightly when pure methane was used as fuel. On the other hand, NO and NO_2 emissions increased up to three times compared to pure methane. Although not detailed for brevity, other studies have also considered different fuel blends, which included combustible and non-combustible components, such as hydrogen [16], ammonia [17] and CO_2 [18], in the combustion process.

Ref. [19] studied not only the efficiency aspects of gas turbines, but also how the use of associated gas instead of natural gas affects their lifespan. The results indicated that for an ambient temperature of 288.15 K, natural gas has higher efficiency compared to associated gas.

In addition, more associated gas is required to produce the same output power in comparison to natural gas. Finally, the lifetime of the blades is approximately 33 thousand hours for associated gas and 49 thousand hours for natural gas. The difference is related to the higher turbine inlet temperatures for turbines operating with associated gas.

Based on the above and knowing that the variation in the content of CO_2 can range from 22 % to 68 % (on a molar basis) in some pre-salt fields according to [20], this work aims, for the first time, to understand how the composition of natural gas/CH_4, which is heavily contaminated with CO_2, along with C_2H_6 and C_3H_8, affects the performance of a gas turbine aimed at GtW systems. The study proposes a thermodynamic model considering the design and the off-design conditions of a gas turbine operated with a variable fuel composition. In addition, a parametric study is performed to evaluate how the content of CO_2 can affect key parameters such as fuel mass flow, exhaust gas mass flow, NO_X and CO emissions, exergy destruction in the components, and energy spent for CO_2 separation.

2. Methodology

The global model, which is a combination of several existing submodels available in the literature, was developed in Python™ [21] and is a sequential calculation routine used to calculate the performance of the GtW cycle, which is detailed in Section 2.1. First, based on the boundary conditions shown in Table 1 and on the volumetric or molar fractions of Table 2, the model calculates the stoichiometric coefficients for the combustion of CH_4, C_2H_6, C_3H_8 and CO_2, which are used to calculate the properties of the fuel, such as density, mass, and calorific value. The energy available from fuel combustion makes it possible to determine all thermodynamic states, mass flow rates, excess air, and partial powers throughout the GtW cycle, which is detailed in Section 2.2. Next, the model addresses the calculation of exergy fluxes, exergy destruction – Section 2.3 –, emission of pollutants, and the maximum energy required for the separation of CO_2, as shown in Sections 2.4 and 2.5, respectively. In Section 2.6, an iterative analysis is performed under partial load conditions, considering all mixtures and variables calculated previously.

2.1. Physical configuration

The representative schematic of a typical GtW Brayton cycle is shown in Fig. 1, which is composed of a compressor, combustion chamber, turbine, generator, and, in the present case, a mixer; the latter ensures that the fuel enters the combustion chamber completely mixed. The operational conditions are as follows: air enters the compressor under ambient conditions (state 1), passes through the compressor, and enters the combustion chamber (state 2); the fuel used in combustion reaches T_f and P_f (state 3), the combustion gases enter the turbine (state 4) and are expanded in the turbine generating electricity (state 5). The main assumptions and boundary conditions used to build the numerical model are shown in Table 1.

As shown in Table 1, air enters the compressor at 298.15 K (T_1) and 1 atm (P_1), while the compression ratio is equal to 16.9, according to the SGT-400 [22]. The electrical power generated is fixed and equal to 12.9 MW for all fuel compositions tested, and the design exhaust temperature is 828.15 K. After preliminary simulations with the boundary conditions, it was determined that the turbine inlet temperature is 1487 K. This temperature was fixed in the part-load model to prevent the maximum blade temperature from being exceeded. Also, the efficiency of the electric generator was set at 0.972, and the combustor is responsible for a pressure drop of 5 % as reported in [14]. The isentropic efficiency values of the compressor and turbine were chosen as 0.831 and 0.901 [23].

The turbine is fed by a mixture containing CH_4, C_2H_6, C_3H_8, and CO_2 with the composition indicated in Table 2. The variation in gas content was considered in relation to the analysis presented by Ref. [20], which evaluated 63 pre-salt natural gas samples. The results indicated that the content of CO_2 reached 68 % (molar basis), while ethane varied between 2 % and 9 %, and propane varied from 1 % to 6 %. Therefore, it was decided to keep the ethane and propane content constant at 4 %, while the CO_2 content varied from 2 % to 42 %. The CO_2 values were chosen because only 4 of the 63 samples had a CO_2 content exceeding 45 %.

Table 1
Referred parameters and boundary conditions used in the thermodynamic model.

Parameter	Description	Value	Unit	Reference
η_{gen}	Electric generator efficiency	0.972	–	[14]
η_t	Turbine isentropic efficiency	0.901	–	[24]
η_c	Compressor isentropic efficiency	0.831	–	[24]
η_{gtc}	Net electrical efficiency	0.349	–	[22]
$\dot{W}_{gtc}$	Gas turbine power output	12.9	MW	[22]
T_1	Ambient temperature	295.15	K	–
P_1	Ambient pressure	0.101	MPa	–
T_5	Turbine outlet temperature	828.15	K	[22]
T_3	Fuel temperature	295.15	K	[14]
P_3	Fuel pressure	0.101	MPa	[14]

Table 2
Fuel composition used in the turbine.

Mixture	F-1	F-2	F-3	F-4	F-5
CH_4	90 %	80 %	70 %	60 %	50 %
C_2H_6	4 %	4 %	4 %	4 %	4 %
C_3H_8	4 %	4 %	4 %	4 %	4 %
CO_2	2 %	12 %	22 %	32 %	42 %
Wo [MJ/Nm^3]	59.02	53.26	48.70	37.36	28.90

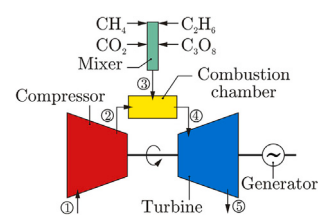

Fig. 1. Representation of the GtW cycle using a gas mixture as fuel.

2.2. Thermodynamic formulation

The Python™ [21]-based numerical model relied on its own libraries (e.g., Numpy and Sympy), third party programs such as Coolprop [25] and Cantera [26], as well as a series of specific models available in the literature to calculate the thermodynamic states and properties for all points listed in Fig. 1. The cycle efficiency and the output power are input data for the model, so it is possible to determine the mass flow rate of the gas required to generate the established electrical power. The values of the isentropic efficiencies were taken from the work of [24].

The thermodynamic model, which is based on the work of [14], starts by determining the equilibrium equations for the combustion of CH_4, C_2H_6, and C_3H_8, which according to [27], can be written as

$$CH_4 + a_0\,(O_2 + 3.76N_2) \rightarrow a_1\,CO_2 + a_2\,H_2O + a_3\,N_2, \tag{1}$$

$$C_2H_6 + b_0\,(O_2 + 3.76N_2) \rightarrow b_1\,CO_2 + b_2\,H_2O + b_3\,N_2, \tag{2}$$

$$C_3H_8 + c_0\,(O_2 + 3.76N_2) \rightarrow c_1\,CO_2 + c_2\,H_2O + c_3\,N_2, \tag{3}$$

where parameters a, b and c are the stoichiometric coefficients. Next, the combustion enthalpies (h_c in J/kg) of each component can be

determined by [27]

$$h_{c|CH_4} = \frac{(a_1 \cdot M_{CO_2} \cdot h_{f|CO_2} + a_2 \cdot M_{H_2O} \cdot h_{f|H2O})}{M_{CH_4}} - \frac{(h_{f|CH_4} \cdot M_{CH_4} + a_0 \cdot M_{O2} \cdot h_{f|O2})}{M_{CH_4}}, \tag{4}$$

$$h_{c|C_2H_6} = \frac{(b_1 \cdot M_{CO_2} \cdot h_{f|CO_2} + b_2 \cdot M_{H_2O} \cdot h_{f|H2O})}{M_{CH_4}} - \frac{(h_{f|C_2H_6} \cdot M_{CH_4} + b_0 \cdot M_{O2} \cdot h_{f|O2})}{M_{C_2H_6}}, \tag{5}$$

$$h_{c|C_3H_8} = \frac{(c_1 \cdot M_{CO_2} \cdot h_{f|CO_2} + c_2 \cdot M_{H_2O} \cdot h_{f|H2O})}{M_{C_3H_8}} - \frac{(h_{f|C_3H_8} \cdot M_{CH_4} + c_0 \cdot M_{O2} \cdot h_{f|O2})}{M_{C_3H_8}}, \tag{6}$$

where M refers to the molar mass in g/mol and h_f is the enthalpy of formation in J/mol for each component. Next, the high (HHV) ([27]) and low (LHV) ([28]) heating values of the mixture, both in J/kg, can be determined as shown below

$$HHV = f_{m|CH_4} \cdot h_{c|CH_4} + f_{m|C_2H_6} \cdot h_{c|C_2H_6} + f_{m|C_3H_8} \cdot h_{c|C_3H_8}, \tag{7}$$

$$LHV = HHV - 2,440 \cdot m_{H_2O}, \tag{8}$$

where f_m is the molar fraction, 2440 refers to the latent heat of vaporization of water in MJ/kg, and m_{H_2O} is the mass of water resulting from combustion in kg. In order to generalize the results, the energy capacity of the fuel mixture is calculated through the Wobbe index (Wo), which, according to [29], measures the energy density of the gas and is used to determine the compatibility limits of the combustion system, and reads as [29]

$$Wo = HHV\left(\frac{\rho_{mix}}{\rho_{air}}\right)^{-1/2}, \tag{9}$$

where ρ_{mix} and ρ_{air} refer to the specific mass of the mixture and air, respectively, both in kg/m^3.

Having determined the energy content of the fuel mixture, one can focus on calculating how the actual cycle would perform, efficiency-wise, for that mixture. Therefore, based on the turbine datasheet selected for the present study [22], i.e., SGT-400, the cycle efficiency reads as follows

$$\eta_{gtc} = 3600/HT, \tag{10}$$

where HT is the heat rate, which is measured in kJ/kWh and is tabulated in [22].

Recall that, as shown in Table 1, the actual power output of the turbine ($\dot{W}_{gtc}$) is fixed at 12.9 MW. Therefore, one must determine the required mass flow rates of the air and the mixture such that the turbine delivers the required power output. The mass flow rate of the fuel mixture is [30]

$$\dot{m}_f = \frac{\dot{W}_{gtc}}{LHV \cdot \eta_{gtc}}. \tag{11}$$

Next, Eqs. (12) and (13) are used to calculate the mass flow rates through the compressor ($\dot{m}_c$) and turbine ($\dot{m}_t$), respectively, measured in kg/s [14],

$$\dot{W}_{gtc} = \eta_{gen} \cdot \left\{\left[\dot{m}_t \cdot (h_4 - h_5)\right] - \left[\dot{m}_c \cdot (h_2 - h_1)\right]\right\} \tag{12}$$

$$\dot{m}_c = \dot{m}_t - \dot{m}_f, \tag{13}$$

where h are the enthalpies at the inlet and outlet of these components evaluated in J/kg. Based on the calculations above, the air/fuel ratio (R_{air}) and the excess air for fuel combustion (ex_{air}), which will be used later to calculate the adiabatic flame temperature, can be determined as follows

$$R_{air} = \frac{\dot{m}_c}{\dot{m}_f}, \tag{14}$$

$$ex_{air} = \frac{R_{air}}{m_{air|st}}. \tag{15}$$

Finally, from a thermodynamic standpoint, one can determine the actual power output contribution of each combustible component of the fuel mixture

$$\dot{W}_{gtc|\mathrm{CH_4}} = \mathrm{LHV}_{CH_4} \cdot \eta_{gtc} \cdot \dot{m}_f, \tag{16}$$

$$\dot{W}_{gtc|\mathrm{C_2H_6}} = \mathrm{LHV}_{\mathrm{C_2H_6}} \cdot \eta_{gtc} \cdot \dot{m}_f, \tag{17}$$

$$\dot{W}_{gtc|\mathrm{C_3H_8}} = \mathrm{LHV}_{\mathrm{C_3H_8}} \cdot \eta_{gtc} \cdot \dot{m}_f. \tag{18}$$

2.3. Exergetic formulation

Next, and through an exergetic analysis [31], it is possible to determine the components and the locations where the exergy destruction is more pronounced in the cycle. Thus, with this analysis, it is possible to determine whether the CO_2 content in the fuel directly affects the exergy destruction rate in the main components of the cycle, i.e., compressor ($\dot{X}_c$), turbine ($\dot{X}_t$) and combustor ($\dot{X}_{fuel}$), which can be written, according to [14], as

$$\dot{X}_c = \dot{m}_c \cdot (x_1 - x_2) + \dot{W}_c, \tag{19}$$

$$\dot{X}_t = \dot{m}_c \cdot (x_4 - x_5) - \dot{W}_t, \tag{20}$$

$$\dot{X}_{comb} = \dot{m}_f \cdot x_f + \dot{m}_f \cdot x_3 + \dot{m}_c \cdot x_2 - \dot{m}_t \cdot (x_{4,f} + x_{4,q}), \tag{21}$$

where $\dot{X}$ is the physical exergy in W, $x_{i,f}$ is the specific physical exergy in J/kg, $x_{i,q}$ is the specific chemical exergy in J/kg, $\dot{W}_c$ is the power spent in the compressor and $\dot{W}_t$ is the power produced in the turbine, both measured in W. In Eq. (21), the exergy destruction rate is given by the exergy balance in the combustor and is a function of physical and chemical exergies, which, according to [31], can be determined as

$$x_i{}^q = \sum y_i \cdot \bar{x}_i + \bar{R} \cdot T_0 \cdot \sum y_i \cdot \ln(y_i) \cdot y_i, \tag{22}$$

$$x_i{}^f = c_p(T_i, P_i) \cdot (T_i - T_0) - T_0 \cdot c_p(T_i, P_i) \cdot \ln(T_i/T_0) + T_0 \cdot R \cdot \ln(P_i/P_0), \tag{23}$$

where y_i is the mole fraction of the exhaust gases, $\bar{x}_i$ is the standard exergy of the components present in the gases in J/mol [31], T_0 is the dead-state temperature in K, T_i is the temperature of the state i in K, P_0 is the dead-state pressure in Pa, P_i is the pressure of the state i in Pa and c_p is the specific heat at constant pressure in J/kg · K.

Finally, exergetic efficiency is given by the ratio between net work and the total amount of energy supplied by fuel [14]

$$\eta_{ex} = \frac{\dot{W}_{gtc}}{\dot{X}_f}. \tag{24}$$

2.4. Emissions formulation

The model chosen to calculate the NO_X and CO emissions follows the formulation proposed by [32], which modifies Lefebvre's correlations to account for the exponential dependence of the reaction rate on the

temperature, as described by [33]. The NO_X and CO emissions, both in g/kg_{fuel}, are, respectively, given as

$$NOx = \frac{0.15 \cdot 10^{16} \cdot (\tau - 0.5 \cdot \tau_{ev})^{0.5} \cdot \exp(-71,000/T_{af})}{P^{0.05} \cdot (\Delta P/P)^{0.5}}, \tag{25}$$

$$CO = \frac{0.18 \cdot 10^{9} \cdot \exp(7800/T_{pz})}{P^{2} \cdot (\tau - 0.4 \cdot \tau_{ev}) \cdot (\Delta P/P)^{0.5}}, \tag{26}$$

where τ, which is measured in s, is referred to as residence time, τ_{ev} is the fuel evaporation time in s, T_{af} is the adiabatic flame temperature in K, P is the inlet pressure of the combustor chamber in Pa, ΔP is the actual pressure drop occurring in the combustion chamber in Pa, T_{pz} is the temperature of the primary zone in K.

It is worth mentioning that the model depends on parameters such as the adiabatic flame temperature and the primary zone temperature. The former temperature was calculated using Cantera [26], while considering the adiabatic flame condition at constant pressure, composition, and excess air. For the latter, it was estimated using a balance between the energy flows at the turbine inlet, $\dot{m}_c \cdot c_p \cdot T_2 + \dot{m}_f \cdot (c_{p,f} \cdot T_f + LHV)$, and at the primary zone outlet, $\dot{m}_t \cdot c_p \cdot T_{pz}$. Furthermore, additional required information data, such as the time spent in the chamber and the normalized time spent, were assumed to be 0.1 and 0, which are values suitable for this type of problem, as described by [33]. However, the calculated values obtained may be seen as qualitative. This is because the main objective is to understand how the gas composition and turbine load influence the emission of NO_X and CO, and not to find the exact values of the load emissions. Yet, precise values of NO_X and CO would require either a detailed numerical model and/or a controlled experimental analysis, both of which are beyond the scope of this study. Nevertheless, the emission trends obtained are still useful as they reveal the impact of the use of contaminated CH_4.

2.5. Separation energetic cost

In a typical offshore well, the gas removed from the well is treated and the existing contaminants, mainly CO_2, are reduced to acceptable levels for commercialization and transport. The excess CO_2 is then reinjected into the well to maintain pressure and reduce the emission of CO_2 into the environment. In GtW platforms, gas with a high impurity content is used as fuel for the on-site energy generation cycle, thus reducing separation costs. Therefore, the model in the current section aims to determine the energy required to separate CO_2 from CH_4. Next, this energy expenditure is compared with the energy made available by the contaminated fuel, and a new efficiency is calculated through Eq. (31).

The model used is based on the work of [34], where the minimum energy required to separate CO_2 is equal to the variation in the Gibbs Free Energy between states 1′ and 2′; state 1′ is given by the Gibbs free energy before separation of the mixture formed by CH_4, C_2H_6, C_3H_8 and CO_2, Eq. (27), as described in [34], while state 2′ is the sum of the Gibbs free energy of CO_2 after separation, Eq. (28), and the Gibbs free energy of the gas without CO_2, Eq. (29). Therefore, the Gibbs free energy of states 1′, 2′ and that of the gas without CO_2, indicated with a subscript *clean*, can be written as [34]

$$\dot{G}_{gas}|_{1'} = \sum_{i=0}^{N} \frac{\dot{m}_f}{\bar{M}} \cdot y_i \cdot G_i(T_{1'}, P_{1'}) + \frac{\dot{m}_f}{\bar{M}} \cdot R \cdot T_{1'} \sum_{i=0}^{N} y_i \cdot \ln(y_i) + \frac{\dot{m}_f}{\bar{M}} \cdot R \cdot T_{1'} \sum_{i=0}^{N} y_i \cdot \ln(\phi_{i(T_{1'},P_{1'})}), \tag{27}$$

$$\dot{G}_{CO_2}|_{2'} = \sum_{i=0}^{N} \frac{\dot{m}_f}{\bar{M}} \cdot y_{CO_2} \cdot G_{CO_2}(T_{2'}, P_{2'}), \tag{28}$$

$$\dot{G}_{clean}|_{2'} = \sum_{i=0}^{N-1} \frac{\dot{m}_f}{\bar{M}} \cdot y_i' \cdot G_i(T_{2'}, P_{2'}) + \frac{\dot{m}_f}{\bar{M}} \cdot R \cdot T_{2'} \sum_{i=0}^{N-1} y_i' \cdot \ln(y_i') + \frac{\dot{m}_f}{\bar{M}} \cdot R \cdot T_{2'} \sum_{i=0}^{N-1} y_i' \ln(\phi_{i(T_{2'},P_{2'})}), \tag{29}$$

i is the number of components, G_i is the Gibbs free energy of each component in J, calculated by $G_i = H_i - T \cdot S_i$, R is the universal gas constant equal to 8.314 J/mol·K, y_i' is the mole fraction of each component i for gas without CO_2 and finally, ϕ is the fugacity coefficient, which was assumed to be zero according to [34]. Therefore, the Gibbs free energy at state '2' reads

$$\Delta\dot{G}_{2'} = (\dot{G}_{CO_2} + \dot{G}_{gaswithoutCO_2})_{2'} - (\dot{G}_{gas})_{1'}. \tag{30}$$

Finally, the exergy efficiency that accounts for the removal of CO_2 is given by

$$\eta_{CO_2} = \frac{\dot{W}_{gtc} - \Delta\dot{G}_{2'}}{\dot{X}_f}. \tag{31}$$

This approach is used as a comparison to standard generation systems, where gas is treated, separated, and transported through pipelines, and high-efficiency GtW systems, where electricity is generated at the extraction site without the need for gas separation.

2.6. Partial-load formulation

According to [14,35], power generation plants do not operate at full load at all times, so this type of installation must be able to operate at loads lower than the nominal. Therefore, this section describes the part-load formulation used, which was presented in [14,35]. The model consists of a set of analytical equations based on the performance of the cycle components, and is applied to a single shaft equipment with constant rotational speed [35]. The equations relate the design parameters to the off-design parameters based on the turbine operating load. The design parameters were obtained using the thermodynamic equation model presented in the previous sub-sections. The main equations will be presented below, while the complete model can be found in detail in [35].

The dimensionless turbine efficiency ratio $\bar{\bar{\eta}}_{gtc}$, which represents the turbine power output ratio between its off-design and design value [14], is given by a fourth-order polynomial that has the turbine operating load ratio ($\bar{\bar{W}}$) as a dependent variable

$$\bar{\bar{\eta}}_{gtc} = (3.18 \cdot \bar{\bar{W}}) - (4.69 \cdot \bar{\bar{W}}^2) + (3.69 \cdot \bar{\bar{W}}^3) - (1.18 \cdot \bar{\bar{W}}^4). \tag{32}$$

Also, the dimensionless fuel flow rate, which is defined as $\bar{\bar{m}}_f = \dot{m}_f/\dot{m}_{f,des}$, where the subscript *des* indicates the properties evaluated at the design value, is written as a function of $\bar{\bar{W}}$,

$$\bar{\bar{m}}_f = 0.288 + 0.624 \cdot \bar{\bar{W}} + 0.088 \cdot \bar{\bar{W}}^2. \tag{33}$$

The dimensionless airflow rate entering the compressor, $\ddot{m}_c$, which represents the relationship between the mass flow rate in the turbine compressor and the design value, is given by

$$\ddot{m}_c = \frac{\dot{m}_c}{\dot{m}_{c,des}} \cdot \frac{P_{1,des}}{P_1} \cdot \left(\frac{T_1}{T_{1,des}}\right)^{1/2}, \tag{34}$$

where T_1 and P_1 are the inlet conditions, which are constant for any load, and equal to 25 °C and 0.101 MPa.

The compressor pressure ratio ($\ddot{\pi}_c$) and isentropic efficiency ratio ($\ddot{\eta}_c$) can be calculated using Eqs. (35) and (36), respectively,

$$\ddot{\pi}_c = c_1 \cdot \ddot{m}_c^2 + c_2 \cdot \ddot{m}_c + c_3, \tag{35}$$

$$\ddot{\eta}_c = [1 - c_4 \cdot (1 - \ddot{n})^2] \cdot \left(\frac{\ddot{n}}{\ddot{m}_c}\right) \cdot \left(2 - \frac{\ddot{n}}{\ddot{m}_c}\right), \tag{36}$$

where $\ddot{n}$ is the rotation rate, while the constants c_1, c_2 and c_3 are given by

$$c_1 = \frac{\ddot{n}}{p \cdot \left[1 - (m/\ddot{n})\right] + [\ddot{n} \cdot (\ddot{n} - m)]^2}, \tag{37}$$

$$c_2 = \frac{p - 2 \cdot m \cdot \ddot{n}^2}{p \cdot \left[1 - (m/\ddot{n})\right] + [\ddot{n} \cdot (\ddot{n} - m)^2]}, \tag{38}$$

$$c_3 = \frac{p \cdot m \cdot \ddot{n} - m^2 \cdot \ddot{n}^3}{p \cdot \left[1 - (m/\ddot{n})\right] + [\ddot{n} \cdot (\ddot{n} - m)^2]}, \tag{39}$$

where the constants c_4, m and p are, respectively, 0.3, 1.06 and 0.36 according to Ref. [35]. It is worth mentioning that the compressor and the turbine are coupled to the same shaft, so $n = n_c = n_t$ and $n_{des} = n_{c,des} = n_{t,des}$. According to Ref. [35], the turbine in constant operation must maintain the following equilibrium conditions

$$\frac{\dot{m}_t}{\dot{m}_{t,des}} = \ddot{\mu} \cdot \frac{\dot{m}_c}{\dot{m}_{c,des}}, \tag{40}$$

$$\ddot{\pi}_t = \ddot{\phi} \cdot \ddot{\pi}_c, \tag{41}$$

where $\ddot{\phi}$ is equal to 1 and $\ddot{\mu}$ is given by a polynomial function that depends on the air and fuel flow rates. For the turbine, the following expressions are applied for the isentropic efficiency ratio and the exhaust gas mass flow rate

$$\ddot{\eta}_t = [(1 - t \cdot (1 - \ddot{n}_t)^2] \cdot \left(\frac{\ddot{n}}{\ddot{m}_t}\right) \cdot \left(2 - \frac{\ddot{n}}{\ddot{m}_t}\right), \tag{42}$$

$$\frac{\dot{m}_t}{\dot{m}_{t,des}} = \alpha \cdot \left(\frac{T_3}{T_{3,des}} \cdot\right)^{1/2} \cdot \left(\frac{{\pi_t}^2 - 1}{{\pi_{t,des}}^2 - 1}\right)^{1/2}, \tag{43}$$

where t is equal to 0.3 and α is given by

$$\alpha = \left[1.4 - 0.4 \cdot \left(\frac{n_t}{n_{t,des}}\right)\right]^{1/2}. \tag{44}$$

2.7. Validation

The validation and certification of the model were done in three stages. First, the results of the model were compared with the literature for a turbine operating exclusively with CH_4 assuming full load. In this case, a comparison was made with the 11, 13, and 15 MW versions of the SGT-400 turbine. For this purpose, the exhaust mass flow values obtained were compared with the data in the turbine catalog [22]. The results, which are shown in Table 3, report a maximum error of 1.93 % for the 15 MW version, indicating that the full load calculations are correct.

In a second validation and certification process, the partial load formulation was tested. In this case, the work of [14] and partial load data provided by [22] were used. The results shown in Fig. 2 present the catalog data and the curves obtained by the proposed model for the fuel (Fig. 2a) and the total mass flow rates (Fig. 2b) under partial conditions. It can be seen that the fuel, i.e., pure CH_4, in this case, and the total exhaust flow rate are in good agreement compared to the turbine catalog data, therefore, also validating the partial load formulation and implementation.

Table 3
Comparison of the exhaust mass flow ($\dot{m}_t$) between the present results and the catalog data reported in [22] for three different power outputs.

Power output [MW]	Ref. [22] $\dot{m}_t$ [kg/s]	Present work $\dot{m}_t$ [kg/s]	Error [%]
10.50	34.20	33.89	0.90
12.90	40.00	40.13	−0.34
14.30	44.50	43.64	1.93

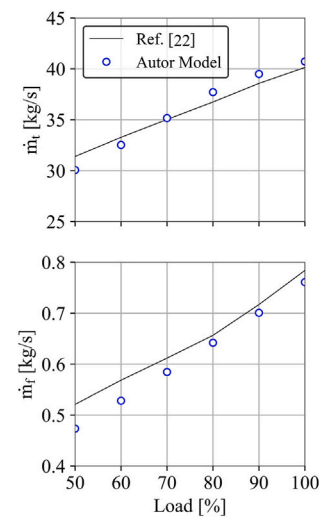

Fig. 2. Validation of the numerical model comparing the present results for mass flow rate of exhaust gases (a) and fuel (b) against [22] for a partial load operation.

The third and final validation process refers to the formulation associated with the cycle's performance while considering a mixed fuel composition. Unfortunately, no data were found that allow one to validate the turbine's behavior. Therefore, in order to ensure the correctness of the mixed fuel calculations, an exergy balance was performed. In this case, Table 4 presents the values obtained by the model for the exergy destruction rate in the compressor ($\dot{X}_c$), in the turbine ($\dot{X}_t$) and in the combustor ($\dot{X}_{comb}$); for the inlet exergy flows in the compressor ($\dot{W}_c$), in the combustion chamber ($\dot{X}_{f,q}$) and ($\dot{X}_{f,f}$); and for the outlet exergy flows ($\dot{X}_5$ and $\dot{W}_t$).

The results show that increasing the content of CO_2 in the fuel mixture increases the exergy destruction in the compressor and turbine, mainly due to the increase in the air and fuel mass flow rates; this will be discussed later in Fig. 3 in detail. In the compressor, since states 1 and 2 remain unchanged, i.e., pressure and temperature are the same for any mixture, the increase in exergy is due to the increase in the work required to compress the air which is caused by the increase in the air mass flow rate, according to Eq. (19). In the turbine, even with an increase in T_5 (Fig. 9) and $\dot{W}_t$, the increase in exergy destruction rate is due to the increase in the mass flow rate of the exhaust gases. However, exergy destruction rate that occurs in the combustion chamber decreases significantly (i.e., approximately by one order of magnitude) as the CO_2 increases from 2 % (F-1) to 42 % (F-2), which can be associated with the fact that exergy in state four increases with an increase in the gas flow rate. Furthermore, the results indicate that the exhaust gases possess high exergy, which increases with increasing the CO_2 content resulting from the higher outlet temperature. Finally, and arguably more importantly, are the data presented in the last column of Table 2. The values show that the exergy balance is equal to zero for all mixtures tested. This balance is given by the difference between the exergy that enters the system and the sum of the exergy that leaves and is destroyed

Table 4
Exergy balance for the five flue compositions shown in Table 2.

Mix.	Loss [MW]			Outlet [MW]		Inlet [MW]		Exergy balance
	$\dot{X}_c$	$\dot{X}_t$	$\dot{X}_{comb}$	$\dot{X}_5$	$\dot{W}_t$	$\dot{X}_{f,q}$	$\dot{W}_c$	
F-1	2.2	1.6	11.8	10.7	30.9	37.7	19.5	0.0
F-2	2.4	2.4	9.1	12.8	32.6	37.7	21.4	0.0
F-3	2.6	3.1	6.5	14.8	34.2	37.8	23.4	0.0
F-4	2.8	3.7	3.9	16.8	35.8	37.8	25.1	0.0
F-5	3.0	4.3	1.6	18.6	37.1	37.9	26.7	0.0

$(\dot{X}_{in} - (\dot{X}_{loss} + \dot{X}_{out}) = 0)$. The balance for a cycle must be equal to zero, that is, it shows that the exergy provided by the fuel is used to generate power, part of it is destroyed in the components, and another part is carried by the exhaust gases that leave the turbine, in addition to showing that the model is mathematically and physically correct.

3. Results

The results obtained with the present model of the GtW cycle will be presented in two distinct sub-sections: according to the Wobbe index (Section 3.1) and the partial load (Section 3.2).

3.1. Wobbe index

The main parameters of the cycle can be shown as functions of the Wobbe index. In fact, turbine manufacturers commercially use this index to disclose the variation range of the fuel used, so this is indeed a parameter of interest in turbine applications where there might be a variation in the fuel composition; in the present study the Wobbe index of the mixtures varied according to Table 2. However, more importantly, a high Wobbe index indicates a low CO_2 content, and vice versa. Also, high CO_2 content is related to combustion instability, decreasing flame temperature and speed, therefore, compromising stability. As the CO_2 content decreases (Wobbe index increases), the flame becomes more stable, preventing flame extinction.

Fig. 3(a) shows the mass flow rates of the exhaust gas ($\dot{m}_t$) and fuel ($\dot{m}_f$) as a function of the Wobbe index (Wo); note that the mass flow rate of the exhaust gases is the sum of the mass flow rates of air and fuel. As can be seen, the five values for the Wobbe index are considered, each indicating one of the compositions shown in Table 2, and are indicated at the upper end of Fig. 3; the lowest Wobbe values represent mixtures with a higher content of CO_2. As expected, both variables decrease as the Wo increases because, for fuels with higher energy density, a lower flow rate is required to achieve the required power output and, consequently, a lower air flow rate will be required for combustion. Since CO_2 is an inert gas, it reduces the energy released during combustion and, therefore, more fuel is required to reach the temperature of 1487 K at the turbine inlet. The results presented in Fig. 3(b), which show the variation of excess air, ex_{air}, confirm the behavior mentioned above, with an increase in the Wobbe index due to a decrease in the excess air used in fuel combustion.

Fig. 4(a) and (b) show the individual contribution of each combustible gas present in the fuel mixture to the total power produced and the exergy destruction rate in the compressor, turbine and combustion chamber, respectively. In Fig. 4(a), it can be seen that the power generated by ethane and propane decreases with increasing the Wobbe index, while the power generated by methane increases with increasing Wo. In other words, since the cycle is set to produce 12.9 MW, regardless of the composition of the fuel mixture, as shown in Table 2, the actual power output share for methane is greater for composition F-1, which has 2 % of CO_2 and 90 % of CH_4, than in composition F-5, which has 50 % of CH_4 and 42 % of CO_2. This is due to fuel consumption, as the overall calorific value of the mixture decreases when the mixture is converted from composition F-1 to F-5. As for exergy, there is an increase in exergy destruction in the compressor and turbine due to the increase in the mass flow rate of air and exhaust gases, which, according

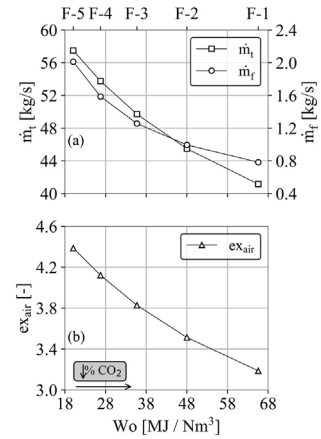

Fig. 3. (a) Variation of mass flow rates of exhaust gases ($\dot{m}_t$) and fuel ($\dot{m}_f$), and (b) the excess air (ex_{air}) as a function of the wobbe index.

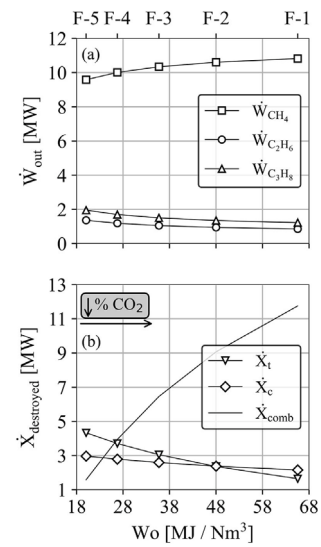

Fig. 4. Partial output power for different fuels (frame a) and the exergy destruction rate (frame b) as a function of the wobbe index.

to Eqs. (19) and (20), show that the higher the content of CO_2 in the fuel (i.e., lower Wo), the greater the exergy destruction due to the increased fuel flow, which, in turn, increases air and exhaust flows. For the combustion chamber, there is a significant increase in exergy destruction rate with an increasing Wobbe index (i.e., lower content of CO_2). This occurs because the combustion chamber outlet temperature is kept constant for all compositions (i.e., 1487 K), and there is an increase in the mass flow rate of the exhaust gases when the Wobbe index decreases. Therefore, exergy in state 4 (i.e., outlet of the combustion chamber) increases, reducing the exergy destruction rate in the combustor, according to the Eq. (23). It can also be noted that the exergy destruction rate is directly related to the composition of the fuel. As CO_2 is an inert gas, a greater amount of gas is burned when comparing the F-5 and F-1 compositions, therefore the exergy destroyed decreases.

The NO_X and CO produced from combustion are shown in Fig. 5(a). It can be seen that, with an increase in the content of CO_2 and, consequently, a decrease in the Wobbe index, the percentage of NO_X per kilogram of fuel decreases drastically. At the same time, there is an increase in CO emissions. This occurs mainly due to the decrease in the adiabatic temperature of the flame, as will be shown later in Fig. 9; this behavior was verified by [13]. It should be mentioned that the influence

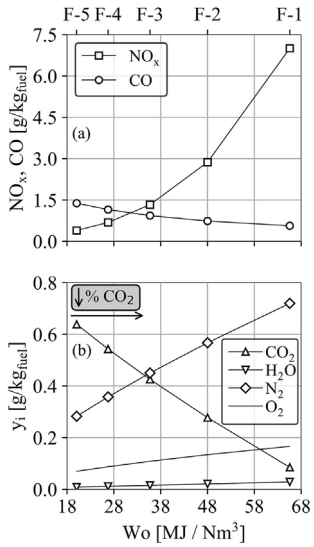

Fig. 5. Mass (a) and molar (b) fractions of exhaust gases as a function of the wobbe index.

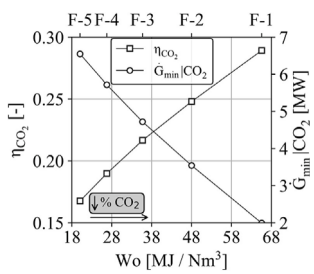

Fig. 6. Effect of the wobbe index on the exergy efficiency after separation (η_{CO_2}) and on the energy required ($\dot{G}_{min}|CO_2$) to separate CO_2 from CH_4.

of fuel composition on NO_X and CO emissions is qualitative, as described earlier in Section 2.4. The other combustion products, i.e., O_2, N_2, and H_2O, which are shown in Fig. 5(b), increase with the increase in the Wobbe index (i.e., a decrease in the content of CO_2). This happens because the increase in the Wobbe index indicates an increase in CH_4 in the fuel and a decrease in CO_2. Consequently, the content of O_2, N_2 and H_2O increases. In contrast, the emission of CO_2 exhibits the opposite behavior since the decrease in the Wobbe index is caused by the increase in the content of CO_2.

As mentioned previously, conventional generation plants separate CO_2 present in the mixture due to transportation restrictions, but this separation generates an energy cost. Thus, Fig. 6 shows the minimum energy required to separate CO_2 from fuel ($\dot{G}_{min}|CO_2$), in addition to the exergy efficiency (η_{CO_2}) of the cycle considering the energy cost ($\dot{G}_{min}|CO_2$) of the separation process. It is worth mentioning that the thermodynamic state before separation was considered at 423.15 K and 1 bar. As expected, the separation energy $\dot{G}_{min}|CO_2$ increases as the Wo decreases. The decrease in Wo indicates that more CO_2 is present in the fuel, increasing the energy required for separation. The decrease in the calculated efficiency η_{CO_2} occurs due to the increase in the energy used to separate CO_2 ($\dot{G}_{min}|CO_2$) as it is considered that part of the cycle output energy is used in the separation process.

Moreover, Fig. 6 raises an interesting question of whether to use fuel with a high CO_2 content or to add a process to separate it from CH_4. According to Fig. 6, there is an increase of approximately 5 MW in the energy spent for separation when comparing gases with CO_2 contents of 2 % and 42 %, that is, the separation process has a high energy cost.

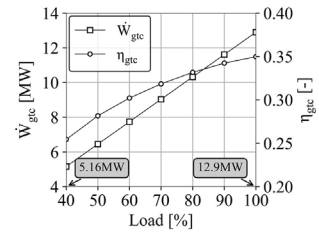

Fig. 7. Output power ($\dot{W}_{gtc}$) and cycle efficiency (η_{gtc}) as a function of load required to the turbine.

Thus, a second turbine operating at 40 % load and using 0.43 kg/s of natural gas (F-1 mixture) would be necessary to generate a total of 12.9 MW with the separation process included. However, the use of contaminated fuel results in an increase in CO_2 emissions (Fig. 5b), but there is a decrease in NO_X emissions (Fig. 5a), hence, indicating the existence of a possible trade-off.

3.2. Partial-load model

Fig. 7 shows the power and efficiency values as a function of the partial load. As expected, both values decrease as the turbine load decreases. For 40 %, the power is 5.16 MW, and the efficiency is 0.254, while for 100 %, the power is 12.9 MW, and the efficiency is 0.349, respectively. As shown in Section 2.6, the cycle efficiency, which is calculated using Eq. (32), and the output power depend on the turbine workload. Thus, both values do not depend on the mixture used, since the output power and the turbine efficiency are input data to the system and feed the model to calculate the other off-design parameters.

In Fig. 8, the values of $\dot{m}_t$ (frame a) and $\dot{m}_f$ (frame b) are plotted as a function of the load. As expected and since the power output is fixed at 12.9 MW, both mass flow rates increase as the CO_2 fraction increases in the fuel mixture. Also, both values increase with the turbine load, regardless of the fuel used. For the mixture with the highest content of CO_2 (F-5), the exhaust gas flow rate and the fuel flow rate vary from 34.5 to 57.4 kg/s and from 1.11 to 2.01 kg/s, respectively, for loads of 40 % and 100 %. This increase is due to the increase in output power and cycle efficiency, as shown in Fig. 7. In both frames (a and b), the curves move away from each other as the turbine load increases. This is a numerical effect of the part-load model. However, this effect correctly reflects the worsening of the cycle efficiency since, for smaller loads, the system requires more air, and consequently the excess air is greater, as described by [14]. This behavior is confirmed later in Fig. 10.

Fig. 9(a) shows that the turbine outlet temperature decreases with increasing load; this behavior is related to the isentropic efficiency of the turbine, which is discussed in Fig. 11. The temperature difference between the turbine inlet and the outlet demonstrates how much mechanical energy is generated through gas expansion. Because the isentropic efficiency is directly linked to the outlet temperature, for high efficiencies, there is better utilization of thermal energy, and consequently the outlet temperature is lower. Therefore, the temperature behavior is easily explained by the decrease in the turbine's isentropic efficiency with decreasing workload. Analyzing Fig. 9(b), it is observed that the adiabatic flame temperature increases with the turbine load, since the adiabatic flame temperature depends on the inverse of the excess air $1/ex_{air}$. Another behavior noted in frame b is that the adiabatic flame temperature decreases with increasing the CO_2 content in the mixture, which was also verified in [13].

The excess air, shown in Fig. 10, exhibits an interesting behavior with fuel composition. It can be observed that excess air increases with an increase in CO_2 content, as seen in Fig. 3(b). This indicates that the reduction in the calorific value of the gas necessitates the use of more fuel, and, consequently, more air is required, increasing the excess air and the air/fuel ratio. Regarding the load, it is noted that the air/fuel

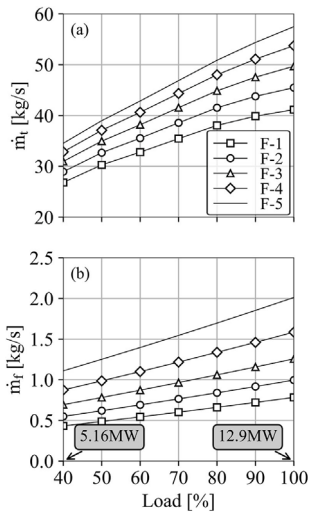

Fig. 8. Mass flow rate of exhaust gases (a) and fuel (b) as a function of partial load for different fuel compositions.

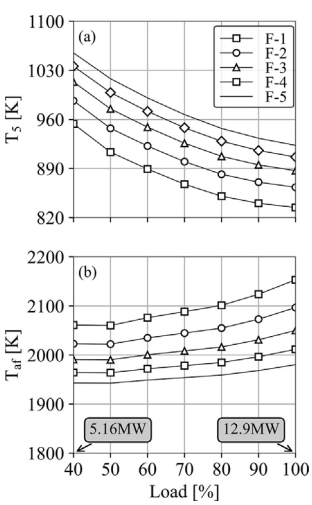

Fig. 9. Effect of the partial load on the turbine outlet (a) and adiabatic flame (b) temperatures.

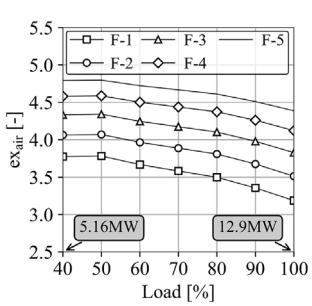

Fig. 10. Excess air as a function of partial load for the five fuel compositions considered.

ratio does not remain constant; that is, the excess air decreases with the increase in the load for all mixtures. This is due to the variation in the value $\dot{\mu}$ present in Eq. (40), which relates the air and fuel flow values for the nominal and off-design conditions. Unlike in [14], where the value of $\dot{\mu}$ was assumed to be equal to 1, in the present work this value was calculated from the data provided by [22].

Fig. 11 shows that the efficiencies of the isentropic turbine and compressor increase with the turbine load and with the increasing CH_4 content in the fuel. At 90 % load, the efficiencies already reach the nominal values of 0.901 and 0.831 for either mixture. As explained previously, the isentropic efficiency of the turbine, Fig. 11(a), is directly associated with the turbine inlet and outlet temperatures. The isentropic efficiency of the compressor, Fig. 11(b), depends on its compression ratio. As the compression ratio decreases, the inlet pressure of the combustion chamber (P_2) decreases, and consequently the temperature (T_2), reducing the isentropic efficiency of the compressor.

NO_X emissions, which are shown in Fig. 12(a), increase with turbine load, reaching a maximum of 7 g/kg$_{fuel}$ for 12.9 MW power, and decrease with increasing CO_2 content, going from a maximum of 7 g/kg$_{fuel}$ to approximately 0.40 g/kg$_{fuel}$. Fig. 12(b) shows that CO has the opposite behavior to NO_X, increasing with increasing CO_2 and decreasing with increasing turbine load. The behavior is explained in Fig. 9(b), which shows that CO and NO_X emissions depend on the adiabatic temperature of the flame. NO_X emissions increase due to thermal NOx (Zeldovich) mechanism [36]. As T_{af} increases, the N_2 present in the air dissociates and reacts with oxygen, and this reaction increases

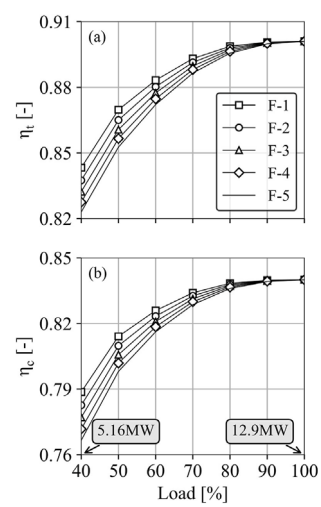

Fig. 11. Isentropic efficiency of the turbine (a) and compressor (b) as a for different partial loads and the compositions reported in Table 2.

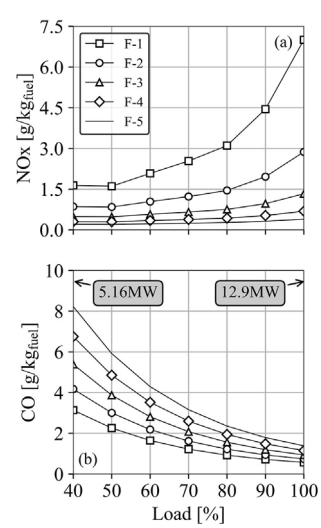

Fig. 12. Variation of the mass fraction of exhaust gases (i.e., CO and NO_X) with the partial load.

exponentially with temperature [37]. As for CO, an increase in T_{af} results in a decrease in emissions, since the oxidation of CO to CO_2 requires high temperatures.

4. Conclusions

With the growing search for energy efficiency and new energy sources, GtW (Gas-to-wire) systems are a plausible compromise solution. In this article, the influence of fuel composition on a Brayton cycle was studied theoretically aiming at oil well applications. The numerical model comprises several analyses, such as the thermodynamics and exergy of the cycle, pollutant emissions, the energy used to separate CO_2, and, finally, the cycle operating at partial load. The study considers the influence of fuel on several cycle parameters, such as the mass flow of exhaust gases, the mass flow of fuel, excess air, the exergy destruction rate in the compressor, turbine, and combustion chamber, pollutant emissions, and isentropic efficiencies.

In short, the results show that both gas and fuel flow rates increase with increasing CO_2 content in the mixture and with increasing turbine workload, a behavior confirmed by data provided by the manufacturer of the SGT-400 turbine [22]. Excess air exhibits a slightly different behavior, as it increases with the CO_2 content (decrease in the Wobbe index) and decreases with turbine load. Regarding pollutants, NO_X and CO emissions decrease and increase with increasing CO_2 content, respectively. However, increasing the turbine load increases NO_X emissions and decreases CO emissions.

It is worth mentioning that the objective of implementing GtW systems is to use as fuel the associated gas with a high contaminant content that, in standard systems, is burned to avoid CH_4 emissions into the atmosphere. One of the main economic advantages of this system is that there is no need to separate CO_2 from the extracted gas. Therefore, one of the results presented concerns the energy required for the separation process. The results presented show that, for fuels with a higher CO_2 content, the use of up to approximately 6.5 MW can be avoided for a fuel flow rate equal to 2.02 kg/s. In other words, despite the reduced cycle efficiency due to the presence of CO_2 mixed with CH_4, increasing the fuel mass flow results in a positive net energy balance, considering that the gas did not go through the separation process, avoiding power expenditures.

In summary, this study shows the implications of running GtW systems with a high level of associated gas. Future studies can further expand this analysis in at least two directions. First, one should improve the model presented by numerically simulating the combustion chamber and all kinetic and geometric details associated with it; this would also improve the accuracy of the associated emissions, which, in the present model, are qualitative. Second, a carbon capture system should be modeled aiming to quantify technically, economically, and environmentally the impact of running a GtW with a clean fuel (i.e., fully separating CO_2) or associated gas (i.e., contaminated with CO_2).

CRediT authorship contribution statement

Ederson Sandrin: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Thaís D. Luz:** Writing – review & editing, Supervision, Conceptualization. **Edson Bazzo:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Alexandre K. da Silva:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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